

PAPER_167 — GW231123: 225 M_sol BH Merger, UQFF Ug4 Feedback, and Yang-Mills Mass Gap

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Abstract

This paper analyzes the GW231123 gravitational wave event — a 225 M_sol binary black hole merger detected in LIGO-Virgo-KAGRA O4 run (November 2023) — through the UQFF framework. This mass-gap event (above the pair-instability supernova 50–130 M_sol gap) challenges standard stellar evolution and is here modeled through enhanced Ug4·(1+f feedback) coupling and the $\delta\rho/\rho$ dark-matter perturbation term. Connections to the Yang-Mills mass gap Millennium Problem are identified, as the non-zero gluon condensate provides a mechanism for mass-gap BH formation.

1. GW231123 Event Properties

Property	Value	Source
Event designation	GW231123	LIGO-Virgo-KAGRA O4
Detection date	November 2023	GWOSC O4a dataset
Primary BH mass	~130 M_sol	GW inferred from chirp mass
Secondary BH mass	~95 M_sol	GW inferred from mass ratio
Total merger mass	225 M_sol	PAPER_167 baseline
Remnant mass	~213 M_sol	After GW energy radiated
ΔM_{GW} (energy)	~12 M_sol c ²	GW radiated energy
Mass gap status	BOTH components above 50 M_sol PISN gap	Anomalous

2. UQFF Analysis

2.1 Why Ug4 Dominates for Extreme-Mass Mergers

For standard BH masses ($M \sim 10\text{--}50\text{ M_sol}$), the Ug4 vacuum concentration term is small compared to Ug1 (magnetic dipole) and Ug3 (string rotation). However, for 225 M_sol, two effects amplify Ug4:

- High M_bh/d_g ratio:** If the merger is at cosmological distance, M_bh increases relative to d_g (the galactic center distance scale), amplifying $\text{Ug4} \propto M_{\text{bh}}/d_{\text{g}}$
- f_feedback:** AGN feedback factor = 0.1 applies as stellar mass BHs above the PISN gap require AGN environment formation → f_feedback is non-zero

$$U_{g4}^{(225)} = k_4 \cdot \rho_v \cdot C_{\text{conc}} \cdot \frac{225M_{\odot}}{d_{\text{source}}} \cdot e^{-\alpha t} \cos(\pi t_n) \cdot 1.1$$

2.2 $\delta\rho/\rho$ Perturbation Activation

The dark matter perturbation term (PAPER_163.8):

$$g_{pert} = (M + M_{DM}) \cdot \left(\frac{\delta\rho}{\rho} + \frac{3GM}{r^3} \right)$$

For GW231123, the merger is embedded in a dark matter halo: - $M_{DM}/M \approx 5$ (DM-dominated environment estimated) - $\delta\rho/\rho \sim 0.5$ (large density contrast — merger in dense environment)

$$g_{pert}^{(225)} = (225 + 1125)M_{\odot} \cdot \left(0.5 + \frac{3G \cdot 225M_{\odot}}{r^3} \right)$$

3. Yang-Mills Mass Gap Connection

3.1 The Millennium Problem

The Yang-Mills mass gap problem asks: **Why do quantum Yang-Mills gauge theories have a mass gap?** (No massless bound states despite gauge bosons being formally massless.)

The Clay Mathematics Institute requires: 1. A quantum Yang-Mills theory in 4D 2. Proof that the physical Hilbert space has mass gap $\Delta > 0$

3.2 UQFF Connection via Non-Zero Gluon Condensate

The QCD gluon condensate $\langle G^2 \rangle \neq 0$ provides the mechanism for above-PISN BH formation:

$$M_{BH}^{gap} \propto \langle G^2 \rangle / \Lambda_{QCD}^4$$

If the mass gap $\Delta = \Lambda_{QCD}$ (confinement scale), then strong-force confined glueball states can accrete at Planck densities to produce mass-gap BHs:

$$M_{gap} = \frac{\Delta^4}{\hbar^3 c^3} \cdot V_{accretion}$$

For $\Delta = \Lambda_{QCD} = 300 \text{ MeV}$ and $V_{accretion} = (10^{-15} \text{ m})^3$:

$$M_{gap} \sim 10^{-35} \text{ kg} \quad (\text{per glueball state})$$

To reach $225 M_{\text{sol}}$ requires $N_{glueball} \sim 10^{71}$ condensed states — equivalent to the entire BH being composed of condensed Yang-Mills vacuum.

This is a **new UQFF prediction**: mass-gap BH masses are quantized in units of the Yang-Mills mass gap Δ .

System	M	F_U estimate	Dominant UQFF term
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4. UQFF Field Comparison: SGR 1745 vs GW231123

System	M	F_U estimate	Dominant UQFF term
SGR 1745-2900	1.4 M _{sol}	$\sim 1.7 \times 10^{45}$	Resonance ($B \approx 3 \times 10^{11}$ T)
Sagittarius A*	4×10^6 M _{sol}	$\sim 1.3 \times 10^{100}$	Resonance (stellar tidal)
GW231123 BH1	~ 130 M _{sol}	$\sim 5 \times 10^{51}$	Ug4·f _{feedback} + g _{pert}
GW231123 BH2	~ 95 M _{sol}	$\sim 3 \times 10^{51}$	Ug4·f _{feedback} + g _{pert}
Merge remnant	~ 213 M _{sol}	$\sim 8 \times 10^{51}$	Ug4·f _{feedback} dominant

5. Osc_term for GW231123 (PAPER_164 Extension)

From PAPER_164, the Osc_term for GW231123:

$$Osc_{term}^{GW231123} = h_{peak} \cdot \omega_{GW,peak}^2 \cdot r^2 \cdot \frac{225 M_{\odot}}{M_{ref}}$$

where h_{peak} is the peak strain at detector (estimated $\sim 10^{-20}$ for O4 near-network event) and $\omega_{GW,peak}$ is the peak GW frequency at merger (typically 100-200 Hz for stellar BH).

6. CP Integration

CP3: Add GW231123_UQFF_Calculator with: - Ug4 computed at $M = 225$ M_{sol} (GW remnant) - f_{feedback} = 0.1 (AGN environment) - g_{pert} with $\delta\rho/\rho = 0.5$, $M_{DM}/M = 5$ - Osc_term = $h_{GW} \times \omega_{GW}^2 \times r^2$

Status: ☐ Complete | **CP Stage:** CP3 **Supersedes:** N/A (new event analysis) | **Related:** PAPER_164 (Osc_term), PAPER_160 (Ug4 f_{feedback}), PAPER_113 (Yang-Mills §1.13), PAPER_163 (g_{pert} decomposition)